Vijayaadhithan Tamilselvan

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Experience

AI/ML Computational Science Analyst Feb 2025 – Present

Accenture

- Raised top-3 reviewer acceptance from 40% to over 80% with rule-based and negative keyword filters in a Google Cloud Platform (GCP) claims-matching pipeline, reaching 85% assignment coverage
- Automated automotive claims extraction with concurrent large language model (LLM) calls and Pandas normalization, adding 3-layer confidence scoring to flag critical issues
- Reduced processing time by 25% for 10,000+ claims per run with Cloud Run and GCP Workflows
- Synchronized BigQuery and a vector index, reducing missed projects from 500 to under 10 per week
- Reduced candidate projects per claim by 40% and lowered LLM input using semantic and lexical search; classified claims and analyzed issue trends with time-series statistics
- Automated source selection by matching PostgreSQL file metadata to queries with lexical search and LLMs
- Benchmarked client and company evidence reaching LLM context under a 1:5 data imbalance, using precision, recall, and LLM judgments to assess source-selection bias
- Improved retrieval with knowledge-graph clustering, normalization, and ontology mapping to the UI

Software Test Engineer (Contract) Oct 2024 – Feb 2025

Juspay

- Automated Unified Payments Interface (UPI) payment tests to verify transaction accuracy
- Compared pixel-intensity histograms to detect visual inconsistencies across application interfaces

Programmer Analyst Dec 2020 – June 2022

Cognizant Technology Solutions

- Validated Databricks product migrations with Python checks to detect inconsistent records
- Automated client-server tests with Java, Selenium, and Tosca to detect application defects

Projects

Querix Search Live: gainr.in

- Built pgvector and BM25 hybrid search with rank fusion and hosted reranking for Gainr listing discovery
- Updated search indexes incrementally, reusing unchanged embeddings to keep Gainr listings current

Conversational Commerce GitHub README

- Built a retrieval-augmented generation (RAG) shopping API with FastAPI and SQLite task state, preserving preferences across budget refinements and product comparisons
- Combined validated DuckDB extract-transform-load (ETL) with PostgreSQL eligibility filters and Qdrant hybrid retrieval to enforce category and price constraints before reranking

Research

Is Twitch Sustainable? Quantifying the Carbon Footprint of Twitch Streaming 2024

Master's thesis, Uppsala University | uu.diva-portal.org | PDF copy

- Combined controlled AWS streaming experiments and Twitch API observations with statistical analysis to estimate carbon emissions and assess streaming conditions

Skills

Languages and data: Python, Java, Pandas, PostgreSQL, SQLite, BigQuery, DuckDB, Databricks, ETL
AI and retrieval: RAG, LLM applications, pgvector, BM25, Qdrant, rank fusion, reranking
Platforms and tools: Google Cloud, Cloud Run, Workflows, FastAPI, Selenium, Tosca

Education

Uppsala University — Master of Science in Data Science, Machine Learning and Statistics 2024

Anna University — Bachelor of Engineering in Computer Science 2020